

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2024/2025

SEPTEMBER EXAMINATION

**UCCD1213/UCCG1213/UGCS3213 FUNDAMENTALS OF DIGITAL MEDIA
TECHNOLOGY**

FRIDAY, 4 OCTOBER 2024

TIME : 9.00 AM – 11.00 AM (2 HOURS)

EXAMINATION FOR BACHELOR DEGREE

Instruction to Candidates :

This question paper consists of **THREE (3)** questions in Section A and **TWO (2)** questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Should a candidate answer more than **ONE (1)** question in **section B**, marks will only be awarded for the **FIRST** question in that section in the order the candidate submits the answers.

Candidates are allowed to use calculator.

Answer questions only in the answer booklet provided.

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Section A [Total: 75 marks]

This section consists of **THREE (3) COMPULSORY** questions.

Q1. Consider the following statement: -

Multimedia application becomes interactive multimedia when a user is given the options of controlling the media elements.

- (a) Provide **FIVE (5)** media elements related to the above statement. (5 marks)
 - (b) List **any FIVE (5)** navigational controls used in an interactive multimedia application. (5 marks)
 - (c) (i) Describe “non-interactive multimedia applications”. (2 marks)
(ii) Explain an example of “non-interactive” multimedia application. (3 marks)
 - (d) Explain **any FIVE (5)** potential drawbacks of using interactive multimedia contents. (10 marks)
- [Total : 25 marks]

- Q2. (a) A multimedia project is a classic software project which is developed by using multimedia technology with the specific aim of information distribution in an entertaining and compelling manner. Multimedia projects require creativity, artistic as well as programming skills.



Figure 2.1 Team Members in a Multimedia Project

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Q2. (a)(Continued)

Figure 2.1 shows the team members in a multimedia project.

- (i) Elaborate **any THREE (3)** factors that determine the number of team members required in a multimedia development project. (9 marks)
 - (ii) Explain **any THREE (3)** reasons that could affect the success of a multimedia development project. (9 marks)
- (b) With an appropriate example, describe why the use of graphics in conveying messages is more effective and better than using text. (7 marks)
- [Total : 25 marks]

Q3. In general, a video or video clip is multiple electronic signals used to generate a steady source of still images, which simulate movement.

- (a) Elaborate the key differences between video and animation. (5 marks)
 - (b) Frames per second (fps) is a common metric in video capture.
 - (i) Describe what you understand by fps. (3 marks)
 - (ii) Discuss why the minimum value for fps is normally set at 15. (3 marks)
 - (c) One of the advantages of digital video is its support for **random access** or non-linear facilities. Using appropriate example, illustrate what you understand by *random access*. (5 marks)
 - (d) Briefly describe **any THREE (3)** reasons why we have to be careful when incorporating video data in our multimedia applications. (9 marks)
- [Total : 25 marks]

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Section B [Total: 25 marks]

This section consists of **TWO (2) OPTIONAL** questions. Answer **ANY ONE (1)** question.

- Q4. (a) MIDI is one of the most important tools for musicians and producers.
- (i) Explain the use of MIDI. (3 marks)
 - (ii) Discuss how are the audio data stored in MIDI different from digital audio. (6 marks)
 - (iii) Calculate the number of bytes required to store a 3 minutes 20 seconds of mono sound at DVD quality with the resolution of 16 bits.

NOTE: Sampling frequency of DVD is 192 kHz (6 marks)

- (b) Inverse Kinematics (IK) is a technique for animating characters that allows you to control the position and orientation of their limbs by specifying the desired end points, such as the hands or feet.
- (i) Elaborate the benefits of using IK for human motion. (5 marks)
 - (ii) Describe the challenges of using IK for human motion. (5 marks)
- [Total : 25 marks]

- Q5. (a) Explain **any THREE (3)** challenges in designing good user interfaces. (9 marks)

(b)

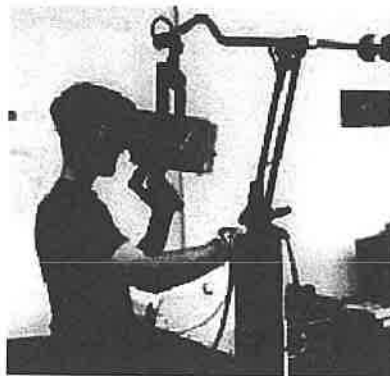


Figure 5.1 Binocular Omni-Orientation Monitor (BOOM)

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Q5. (b)(Continued)

Figure 5.1 shows the Binocular Omni-Orientation Monitor (BOOM), which is a Virtual Reality (VR) environment consisting of a head-coupled stereoscopic display device. Screens and optical system are housed in a box that is attached to a multi-link arm.

- (i) Briefly explain **TWO (2)** advantages of BOOM as compared to Head-mounted Display (HMD). (4 marks)
 - (ii) Explain whether BOOM is classified as immersive or non-immersive VR. (3 marks)
- (c) Elaborate **THREE (3)** differences between VR and Mixed Reality. (9 marks)
[Total : 25 marks]
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